# Staff answers for hazard records

Tuesday 2 March 2027 | P2 | 40 minutes | Year A

Humanities | LAUNCH | Spring 2 Week 2 | HUM-LAUNCH-S2-W02

## Recent records

Use BGS values consistently: A 25 October 2026 at 02:09:23.8 UTC, Bonin Islands Japan, magnitude 6.5, depth 492 km, latitude 27.795 N and longitude 139.618 E. B 24 October at 02:31:30.5 UTC, Vanuatu, magnitude 6.9, depth 10 km, 14.056 S and 165.546 E. C 9 October at 17:56:10.1 UTC, Panama, magnitude 7.7, depth 12 km, 7.587 N and 80.769 W. Do not mix solutions from different catalogues.

## Comparison and plotting answers

A is newest and deepest. C has the largest recent magnitude. B is shallowest. C minus A is a difference of 1.2 in the magnitude values, not an energy multiplier. A minus B is 482 km. Plot A 28 N 140 E, B 14 S 166 E and C 8 N 81 W. Accept sensible approximate points between 30-degree grid lines; check hemisphere and relative position.

## Historic comparison

H is separate: 11 March 2011 at 05:46:24 UTC, Tohoku region Japan, magnitude 9.1, depth 29 km, latitude 38.297 N and longitude 142.373 E. These are USGS values. Do not replace them with a different catalogue solution within the pack.

## Limits and route judgements

Magnitude alone cannot rank impacts. A depth is not a coordinate. Newest does not mean most dangerous. Three selected records cannot establish a trend or exact earthquake prediction. Supported learners may use quiet reading and the worked plot; all routes still need own A–C plots, a date comparison, magnitude/depth evidence and a valid limit. Aspire adds calculations and sample limits. A copied partner answer does not establish understanding.

## Sensitive response

Keep the discussion on values and original diagrams. Do not add disaster footage, personal prompts or graphic accounts. Use the agreed supervised pause and private re-entry. Follow the normal safeguarding route if a concern is disclosed.

Hinge: Which claim does the table support? A: C has the largest magnitude, so it proves C caused most damage. B: B is shallowest at 10 km, but this table alone cannot rank damage. C: A has a depth of 492 km, so it is at 492 degrees north. Answer B.

Parallel check: A is the newest and deepest of the three recent records. Does newest mean most dangerous? Answer: no. The date tells when the event began. A claim about risk or impacts needs other evidence. Ask each learner for a field, unit and limit.

Offer to the Communication portfolio. Keep the learner's own idea, an actual adult audience reply, one NS and the learner's response in the same colour, on the same page or usual mode before R. E follows actual evidence banking. Record actual help. A partner's answer cannot establish both learners' understanding.

Actual observation _____ Help _____
Gap and planned follow-up _____ Actual banking _____